



COMPUTER VISION

TOPICS

Ngô Quốc Việt-2024

Topics

1. Introduction to Computer Vision
2. Camera models
3. Object recognition with global features
4. Object recognition with local features
5. Object classification
6. Motion and optical flows

Introduction to Computer Vision

1. Introduction to computer vision
2. Some main challenges of CV
3. Some applications of CV
4. Getting started with simple installation/implementation

Camera models

1. Pinhole camera model
2. Camera and calibration
3. Perpestive projection
4. Image formation
5. Color and lights

Object recognition with global features

1. Global features in CV

- Histogram
- Histogram backprojection

2. Recognition using global features

- Histogram matching
- Histogram backprojection
- Multi-dimension histogram
- Color derivatives

3. Project: I brand/label recognition in image/video

Object recognition with local features

1. Local feature detector and descriptor
 - Local feature matching
 - Linear transformation and Affine Estimation
 - Homography Estimation
2. Recognition using local features
3. Project 2: panorama image generation
4. Project: 3 face detection in image/video

Object classification

1. Slide window approach
2. Some classification engines
 - Support Vector Machine
 - Artificial neural network or Boosting algorithms.
3. HOG detector
4. Project 4: walking human detection in image/video

Motion and optical flows

1. Motion

2. Optical flows

- Light Invariant Constraint
- The problems of lens
- Lucas-Kanade

3. Project 5: human action recognition in video

References

1. Richard Szeliski. Computer Vision: Algorithms and Applications, 2nd. Edition Springer, 2022.
2. David Forsyth, Jean Ponce. Computer Vision: A Modern Approach 2nd Edition. Pearson, 2012.

Schedule

#Class	Topics	#Lab	Class/Home work	Book Chapter
1	Introduction + Pinhole			
2	Pinhole + Projection	2, 3	Image Formation+ Calibration (opt.)	#1. Chap. 2
3	Global Features	4, 5	Recognition with Histogram Histogram Back Projection;	#1. Chap 3
4	Local Features	6, 7	Haar for face detection	#1. Chap 6
5	Classification	8, 9	SIFT descriptors; Recognition with SIFT; <i>Project: Human detection with SIFT, Haar</i>	#1. Chap 6 #1. Chap 7
6	Motion & Optical Flow	10, 11	Affine + Homography Estimation; <i>Project: Panorama</i>	#1. Chap 8
7		12, 13	Lucas-Kanade Optical Flow; Velocity Estimation/Object Tracking <i>Project: Action recognition</i>	#1. Chap 9 #1. Chap 11